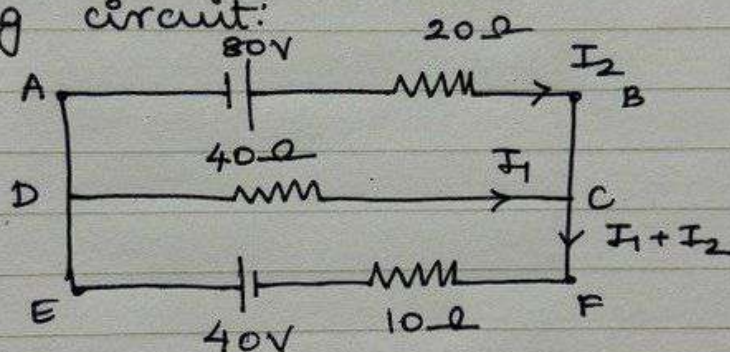


- ① An equilateral glass prism has a refractive index 1.6 in air. Calculate the angle of the minimum deviation of the prism, when kept in a medium of refractive index $\frac{4\sqrt{2}}{5}$.
- ② Why is the wave theory of electromagnetic radiation not able to explain photoelectric effect? How does the photon picture resolve this problem?
- ③ Using Kirchhoff's rules, calculate the current through the 40Ω and 20Ω resistors in the following circuit:



- ④ Derive the expression for the torque acting on a current carrying loop placed in a magnetic field.
- ⑤ Write three points of differences between para, dia and ferro-magnetic materials, giving one example for each.
- ⑥ The magnetic susceptibility of magnesium at 300K is 1.2×10^{-5} . At what temperature will its magnetic susceptibility become 1.44×10^{-5} ?